

Elsherif Adel Abdelhamid

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PROFESSIONAL SUMMARY

Computer and Systems Engineering student specializing in front-end development and AI integration. Experienced in building scalable, intelligent web applications that combine modern UI design with machine learning and automation. Focused on performance, usability, and innovation across both client-side and AI-powered systems.

WORK EXPERIENCE

National Telecommunication Institute (NTI) - Full Stack

Jan 2026

- Soft and Professional Skills
- Full-Stack Web Development
- Database Management and DevOps Deployment

Education

Bsc. in Computer and Systems Engineering

Oct 2020 – Jul 2025

Helwan University, Cairo, Egypt

- **Grade:** Very Good with honor
- **Relevant Coursework:** AI/ML, Database, Embedded Systems, Networking, OS, Data Structures, Computer Vision, OOP

Technical Skills

- **Programming Languages:** HTML, CSS, JavaScript, Python, C/C++, Java, VHDL
- **AI/ML Technologies:** TensorFlow, Keras, PyTorch, OpenAI Gym, Neural Networks, CNNs, RNNs, NLP, Computer Vision
- **Embedded Systems:** STM32, ESP32, AVR, RTOS, I2C, SPI, UART, CAN, MQTT, RFID, Bluetooth
- **Tools & Frameworks:** Git, OpenCV, SQL, Node.js, TinyML

Soft Skills

Problem Solving, Teamwork, Communication, Time Management, Adaptability, Attention to Detail, Continuous Learning.

Courses & Certifications

Al-Madrassa.com - Full Stack

Sep 2025

- 4-months covering HTML, CSS, JavaScript, and OOP principles

Deep Learning Specialization - Coursera (Andrew Ng)

Jan 2025

- 5-course specialization: neural networks, optimization, deep learning

Machine Learning Specialization - Coursera

Nov 2024

- 3-course series covering fundamental ML concepts

Advanced Embedded Systems (ARM, STM32, RTOS)

Sep 2024

- 9-month intensive program by Engineer Amgad Samir

Basic Embedded Systems (AVR)

Sep 2023

- 5-month program by Engineer Mohamed Tarek, and 162 hours by ITI

Projects

AI-Powered Wearable Museum Tour Guide • ESP32, TensorFlow Lite, BLE, LLM

- Designed wearable device with ESP32, KWS for voice activation, BLE for proximity detection
- Created museum staff web interface for content and exhibit management

Personal Portfolio Website • *HTML, CSS, JavaScript* • [GitHub Repo](#) • [Live Demo](#)

- Built responsive portfolio with theme switching, navigation, and social platform integration

Hospital Management System • *Python, SQL* • [GitHub Repo](#)

- Developed comprehensive system with patient/staff/admin workflows and role-based access

NSL-KDD Network Security ML Model • *Python, TensorFlow* • [GitHub Repo](#)

- Applied feature selection using GA, ACO, PSO; compared ML (SVM, RF, KNN) with DL models

Static Malware Analysis Tool • *Python, YARA* • [GitHub Repo](#)

- Extracted macros from malicious documents, identified patterns using YARA rules and VirusTotal API

Keyword Spotting Using TinyML • *C++, TensorFlow Lite* • [GitHub Repo](#)

- Developed voice command recognition with ESP32, SPH0645 microphone, edge AI using TF Lite Micro

Movie Recommender System • *Python, Keras* • [GitHub Repo](#)

- Built content-based recommendation engine using neural networks with user/movie features